

5.5.9

EE25BTECH11057 - Tammineedi Rushil Shanmukha Srinivas

Question: Using elementary row transformations find the inverse of the matrix

$$\begin{pmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{pmatrix} \quad (1)$$

Solution: Given,

$$\mathbf{A} = \begin{pmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{pmatrix} \quad (2)$$

Forming the Augmented Matrix

$$\left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array} \right) \quad (3)$$

Applying elementary row operations to find the inverse

$$\left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3 \rightarrow 3R_3 - 4R_2} \left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ -8 & 0 & 3 & 0 & -4 & 3 \end{array} \right) \xrightarrow{R_3 \rightarrow 3R_3 + 8R_1} \left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \quad (4)$$

$$\left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \xrightarrow{R_2 \rightarrow 3R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 0 & 9 & 2 & -2 & 3 & 0 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_3} \left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 0 & 9 & 0 & -18 & 27 & -18 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \quad (5)$$

$$\left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 0 & 9 & 0 & -18 & 27 & -18 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \xrightarrow{R_2 \rightarrow \frac{R_2}{9}} \left(\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + R_3} \left(\begin{array}{ccc|ccc} 3 & 0 & 0 & 9 & -12 & 9 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \quad (6)$$

$$\left(\begin{array}{ccc|ccc} 3 & 0 & 0 & 9 & -12 & 9 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{R_1}{3}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -4 & 3 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array} \right) \quad (7)$$

The right side part of the Augmented Matrix is $\mathbf{A}^{-1}$

$$\mathbf{A}^{-1} = \begin{pmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{pmatrix} \quad (8)$$